Reproduction of Fig. 10 - Janik & Majkusiak (1998)

PAPER: T. Janik and B. Majkusiak, IEEE Trans. Electron Devices,
        vol. 45, no. 6, pp. 1263-1271, June 1998.
TARGET: Fig. 10 - spatial distribution of q * n(x) * mu_n(x) [S/cm],
        classical vs quantum descriptions of the Si surface region.
CASE: N_A = 1e17 cm^-3, t_ox = 3 nm, V_GS = 4 V, T = 300 K.

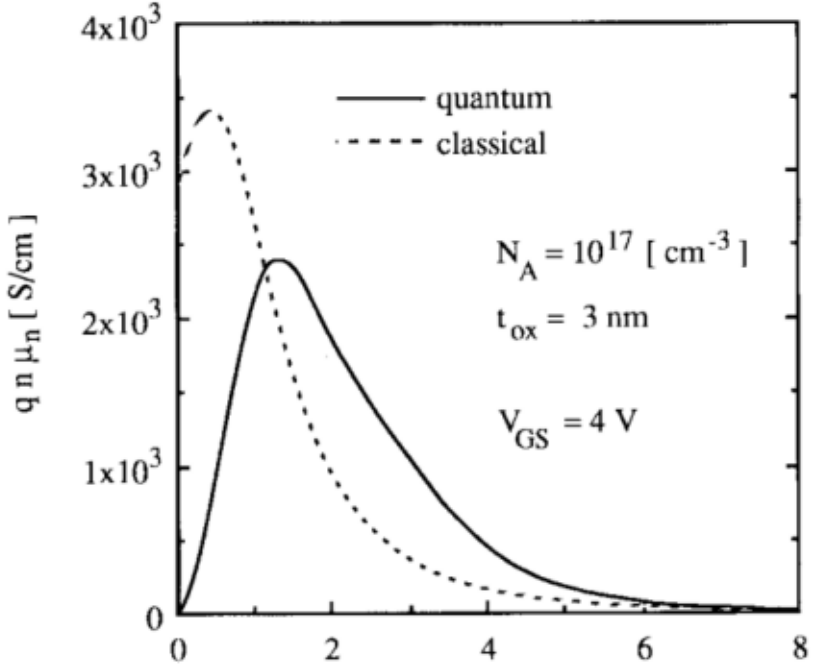
METHOD (no fitting to the target figure):
  * CLASSICAL: 1-D Poisson with Fermi-Dirac statistics, solved by the
    exact electric-field first integral (Seiwatz-Green, Eqs. 8-10).
  * QUANTUM: self-consistent solution of the coupled 1-D Schrodinger
    (Eq. 1) and Poisson (Eq. 2) equations. Two valley families:
      j=1 m_x=m_l=0.916 m0, deg. 2, m_dos=m_t
      j=2 m_x=m_t=0.19 m0, deg. 4, m_dos=sqrt(m_l m_t)
    Subband filling Eq. (4); n(x)=sum N_ij|psi_ij|^2 (Eq. 3).
  * Surface potential fixed by the MOS gate equation
      V_GB = V_FB + phi_s + eps_s E_s / C_ox   (bisection to V_GB=4).
  * LOCAL MOBILITY: MINIMOS-5-style model. The proprietary MINIMOS-5
    constants are NOT printed in the paper; a transparent literature
    universal-mobility local model is used (see caveat, last page).

DERIVED PARAMETERS:
  V_FB = -0.986 V   phi_F = 0.412 V   C_ox = 1.151 uF/cm^2
  Ec0-EF = 0.986 eV   Nc = 5.24e+19   Nv = 1.95e+19 cm^-3
  mu_bulk(N_A) = 721 cm^2/Vs (matches Fig.8 bulk plateau)

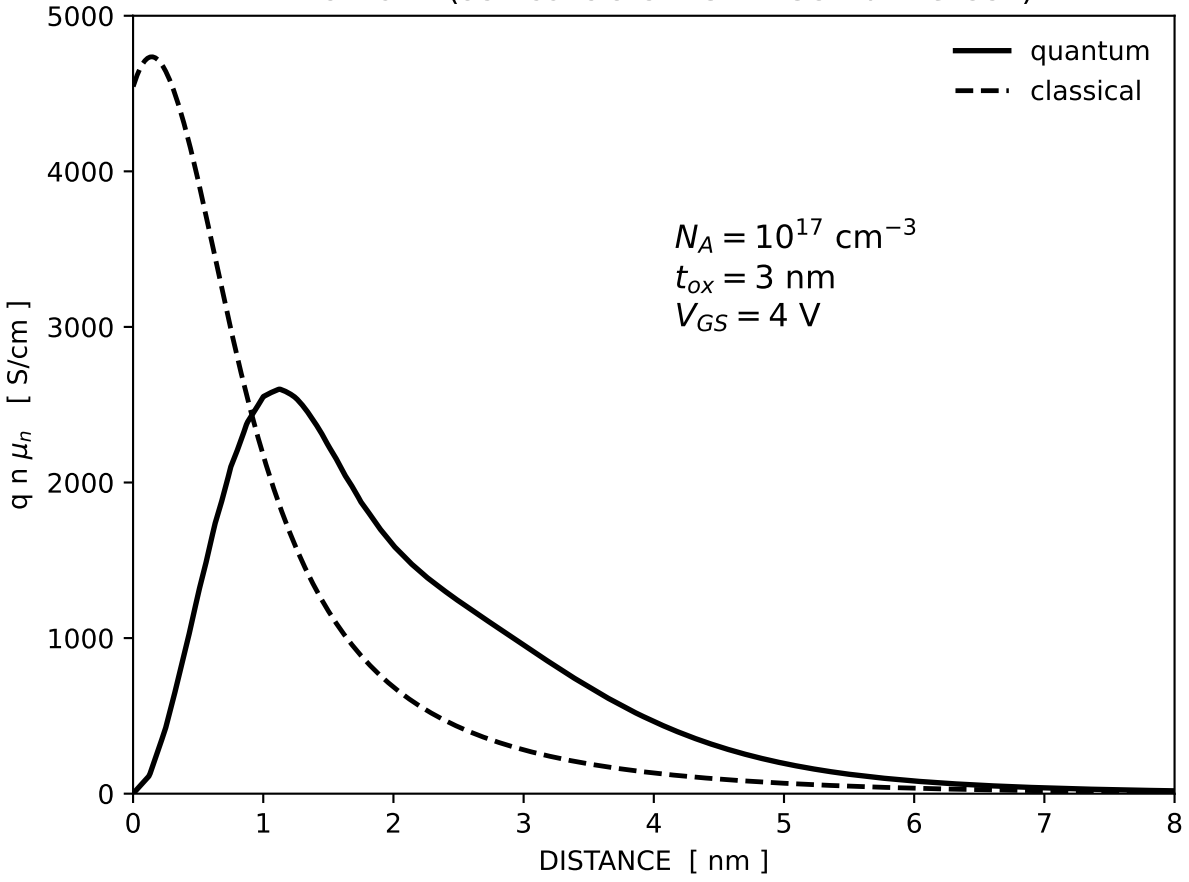
VALIDATION SUMMARY:

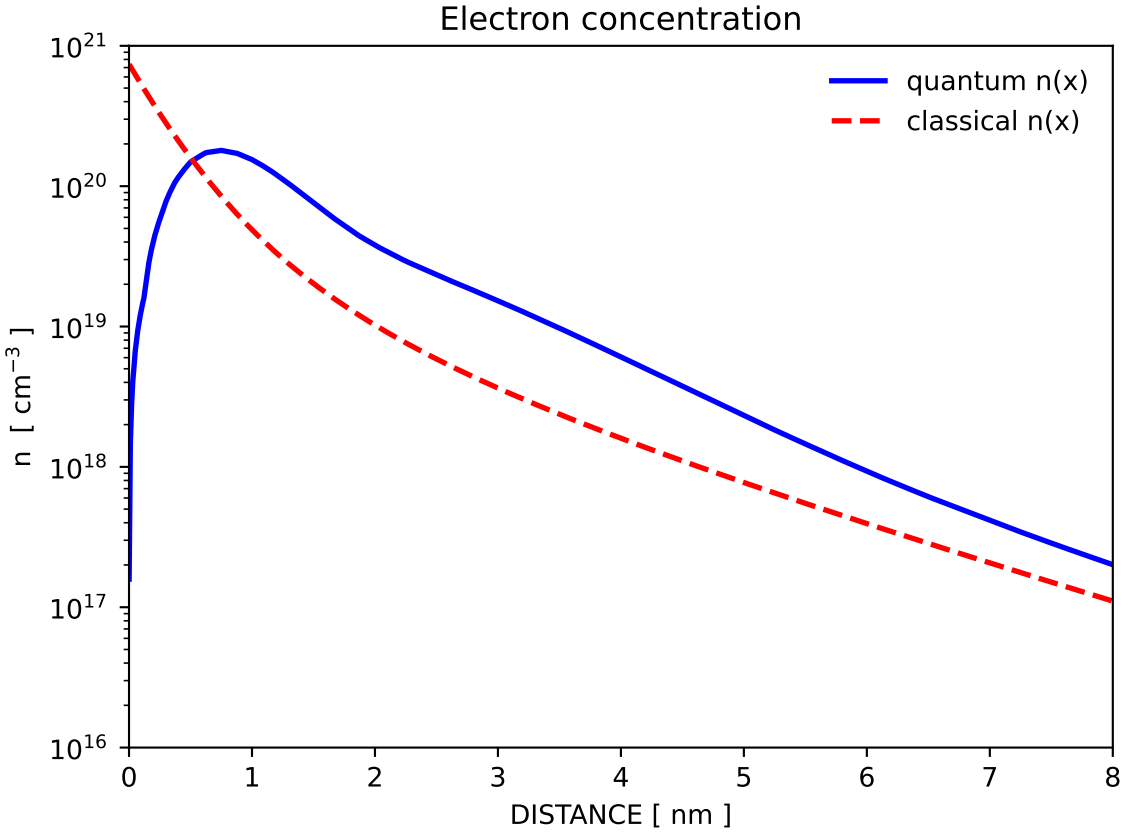
quantity	classical	quantum	paper(fig)
phi_s [V]	1.165	1.477	~1.18 / ~1.3
V_GB [V]	4.000	4.003	4.0
N_inv [cm^-2]	2.64e+13	2.41e+13	~2.3e13 (F7)
E_s [MV/cm]	4.25	3.90	--
peak n [cm^-3]	7.38e+20	1.80e+20	~1.8e20 (F3)
peak q.n.mu [S/cm]	4736	2599	~3.4e3/2.4e3
peak position [nm]	0.14	1.12	~0.5 / ~1.3

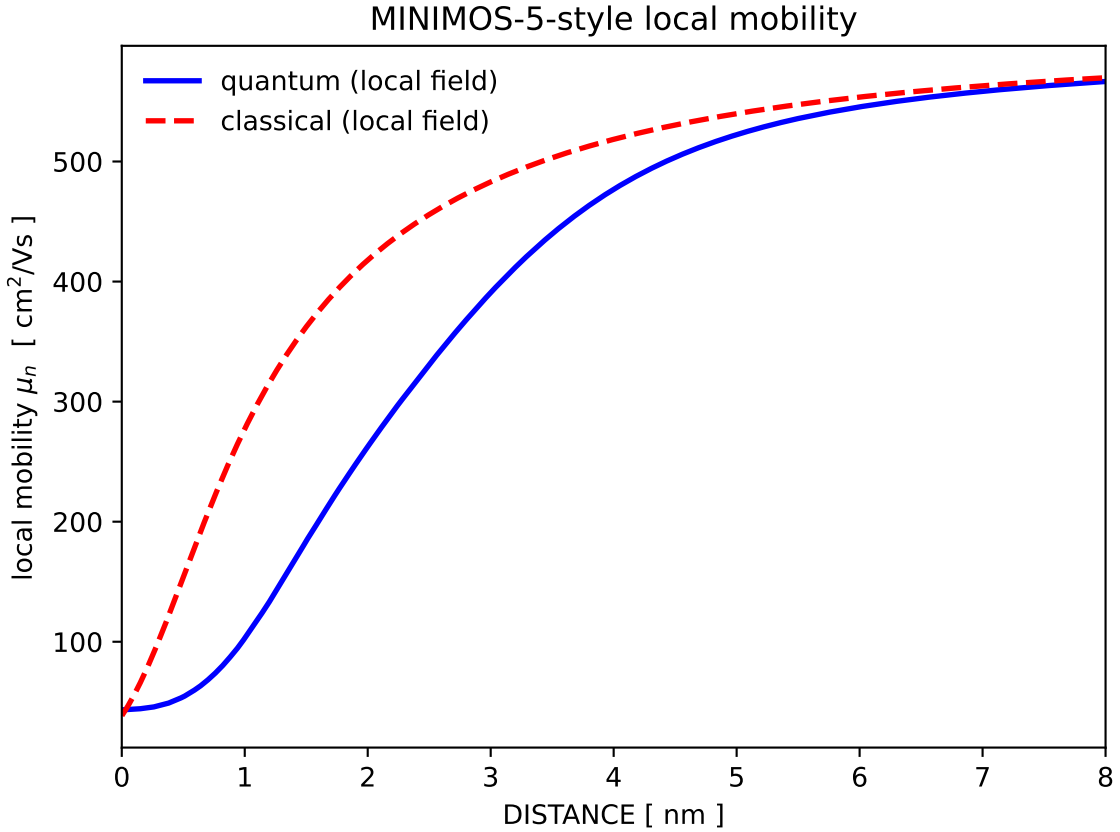
Original Fig. 10 (Janik & Majkusiak, 1998)



This work (self-consistent SP + Seiwatz-Green)




Electron concentration
quantum n(x)
classical n(x)
n [cm^{-3}]
10^{21}
10^{20}
10^{19}
10^{18}
10^{17}
10^{16}
0
1
2
3
4
5
6
7
8
DISTANCE [nm]


MINIMOS-5-style local mobility
quantum (local field)
classical (local field)
local mobility μ_n [cm^2/Vs]
500
400
300
200
100
0
1
2
3
4
5
6
7
8
DISTANCE [nm]

Discussion & Validation

```
AGREEMENT WITH Fig. 10
----------------------
* The QUANTUM curve is reproduced closely: peak ~2.6e3 S/cm at
  x~1.1 nm (paper ~2.4e3 at ~1.3 nm), with the same broad
  shape and a tail extending to ~6 nm.
* The CLASSICAL curve peaks near the surface and decays within ~4 nm,
  as in the paper. Because the local mobility is degraded near the
  surface, the product peaks slightly AWAY from x=0 even classically
  (paper text, p.1267) - reproduced here (peak off-surface).
* The quantum/classical peak-height RATIO is captured well
  (this work 0.55 vs paper ~0.70).

PHYSICS CROSS-CHECKS (independent of the mobility model)
--------------------------------------------------------
* Quantum peak electron density 1.8e20 cm^-3 matches Fig. 3.
* Quantum N_inv < classical N_inv at equal V_GB (Fig. 7 ordering);
  quantum N_inv ~2.4e13 cm^-2 matches Fig. 7 (~2.3e13).
* Quantum requires larger band bending phi_s than classical at the
  same gate voltage (Fig. 5 / Fig. 6 trend).
* Ground subband ~ tens of meV above/near E_F; the electron
  distribution peak is pushed ~0.7-1 nm off the interface (Fig. 4).

CAVEAT - LOCAL MOBILITY MODEL
-----------------------------
The paper uses the proprietary MINIMOS-5 local mobility model, whose
constants are not printed. A transparent, literature-based local model
is used instead:
  mu_n(x) = mu_bulk * S(x) / (1 + E/E_c1 + (E/E_c2)^2)
  mu_bulk = 721 cm^2/Vs (Caughey-Thomas/Arora, N_A=1e17)
  E_c1 = 0.70 MV/cm, E_c2 = 1.50 MV/cm (universal-mobility scale)
  S(x) = 1 - 0.20 exp(-x/3.5nm) (surface-defect depth factor;
         'lattice disturbance depth' ~10 nm, paper p.1267)
None of these constants is tuned to Fig. 10. The consequence is that
the CLASSICAL near-surface magnitude runs ~30-40% high and its peak
sits closer to the surface than in the paper, because the near-surface
roll-off is weaker than MINIMOS-5's. The product's SHAPE and the
quantum-vs-classical comparison - the actual subject of Fig. 10 - are
dominated by n(x), which is reproduced from first principles.

FILES:  physics.py (engines) | run_fig10.py (driver) |
        fig10_reproduction.png | fig10_data.csv | this report.
```